AI-based Pneumonia Detection with Severity Analysis and Grad-CAM

Submitted in the partial fulfillment of the requirements

for the degree of B.Tech in Computer Engineering

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CERTIFICATE

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Abstract

This paper centers around the implementation of an AI model that uses deep learning algorithms to identify pneumonia in X-ray images of the chest. Pneumonia is an important respiratory disorder and needs accurate and timely detection.

Manual identification techniques involve a lot of errors and time consumption. To overcome this problem, Convolutional Neural Network models like ResNet18, DenseNet121, and VGG16 have been used to facilitate automatic classification.

In addition to the identification task, severity analysis and Grad-CAM visualizations are also included in order to make the prediction more comprehensible. It has been shown that this technique produces accurate results.

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Chapter 1

Introduction

1.1 Overview

In recent times, there has been a drastic improvement in the detection and diagnosis of diseases with the help of medical imaging technologies. Out of the several imaging techniques available, the chest X-ray continues to be one of the most popular and cost-effective procedures used in detecting diseases associated with the lungs, such as pneumonia.

Pneumonia is an extremely dangerous respiratory illness that kills thousands of people annually around the world and requires prompt treatment upon diagnosis. The disease is normally diagnosed using X-ray images interpreted by radiologists. Unfortunately, this procedure can sometimes prove to be cumbersome and prone to errors due to human intervention.

Thanks to the recent advancements in Artificial Intelligence, especially deep learning, there has been an influx of intelligent machines capable of analyzing medical imaging data. CNNs have proven to be exceptionally effective in performing classification tasks on images.

1.2 Motivation

The project is motivated by the rise in cases of pneumonia and the need for early and accurate detection. In many healthcare settings specially in rural or resource-constrained environments there is a shortage of experienced radiologists which delays the diagnosis and treatment.

Most existing systems focus only on detecting the presence of pneumonia without providing additional insights such as severity or interpretability. Moreover the medical professionals often require visual explanations to trust AI based predictions.

An intelligent system that can accurately identify pneumonia cases would be very useful, but also capable of detecting more severe complications. It explains severity of symptoms and illness, and provides an explanatory image or animation making better clinical decisions.

1.3 Problem Statement and Objectives

Problem Statement:

Pneumonia is a critical disease that requires timely and accurate diagnosis. Traditional diagnostic methods using chest X-rays depend heavily on expert interpretation, making the process time consuming and prone to human error. Also the existing automated systems often lack understandability and do not provide severity assessment which limits their practical usability in real world.

Objectives:

- Develop a deep learning-based system capable of detecting pneumonia from chest X-ray images.
- Train and evaluate models like ResNet18, DenseNet121, and VGG16.
- Compare model performance using evaluation metrics like accuracy, precision, recall and F1 score.
- Implement severity analysis based on model prediction confidence.
- Apply Grad-CAM to provide visual explanations of model predictions.

1.4 Organization of the Report

The structure of this report is organized as follows.

Chapter 2 presents a view of existing pneumonia detection techniques and their limitations. Chapter 3 describes the proposed system architecture and methodology. Chapter 4 tells about our implementation details and experimental results, and performance analysis. Finally, Chapter 5 concludes the report and outlines future scope.

Chapter 2

Literature Survey

Pneumonia detection using chest X-ray images has been studied for a long time in medical image analysis. The main reason behind this is the need for early and accurate diagnosis of lung-related diseases. Over the years, different techniques have been used, starting from basic image processing methods to more advanced deep learning approaches.

In earlier systems, pneumonia detection was mainly done using traditional machine learning techniques. These methods required manual feature extraction, where features like texture, shape, and intensity were taken from X-ray images. Although these approaches were simple and easy to understand, we found that they were not very effective in capturing complex patterns present in medical data.

The development of deep learning, like Convolutional Neural Network (CNN) the approach changed overall. CNN models are capable of learning features automatically from images, which reduces the need for manual feature engineering. Many research works have shown that deep learning models perform better than traditional methods in medical image analysis [1, 2].

Different CNN architectures have been used in previous studies. Models like ResNet, DenseNet, and VGG have shown good results in image classification tasks. ResNet uses residual connections, which help in training deeper networks. DenseNet connects each layer with every other layer, improving feature reuse. VGG, on the other hand, uses a simple and uniform architecture. Based on what we studied, these models have been widely used for pneumonia detection and have achieved good performance [3, 4, 5].

Another important area that researchers have focused on is interpretability. In medical applications, it is important not only to get predictions but also to understand how the model

is making decisions. Techniques like Grad-CAM help in visualizing important regions in the X-ray image that influence the prediction. This makes the system more transparent and easier for doctors to trust [6].

We also observed that data preprocessing and augmentation play a very important role. Since medical datasets are often limited, techniques like rotation, flipping, and normalization are used to improve model performance. Transfer learning is also commonly used, where pre-trained models are fine-tuned on medical datasets.

2.1 Survey of Existing System

From the literature, we can say that existing pneumonia detection systems fall into two main categories: traditional machine learning-based systems and deep learning-based systems. Traditional methods depend on handcrafted features and generally give lower accuracy.

Deep learning-based systems, especially CNN based models, have shown much better performance. These models automatically learn features from images and can handle variations more effectively. However, most of these systems focus only on classification, i.e., whether pneumonia is present or not.

2.2 Limitations of Existing System or Research Gap

Even though there has been a lot of improvement in this field, there are still some limitations. One major issue is that most models only provide classification results and do not give any information about the severity of the disease. In real medical situations, severity plays a very important role in treatment decisions.

Another problem is the lack of interpretability. Many deep learning models act like black boxes, which makes it difficult for doctors to fully trust the predictions without proper explanation.

We also noticed that the performance of these models depends heavily on the dataset. If the dataset is small or not properly balanced, the accuracy can decrease and the model may not generalize well.

Because of these limitations, there is a need for a better system that can not only detect pneumonia but also provide severity analysis and explanation. In our project, we try to ad-

dress these issues by combining deep learning models with severity scoring and Grad-CAM visualization.

Chapter 3

Proposed System

3.1 Problem Statement

Pneumonia is a serious respiratory disease that requires an early and accurate diagnosis. Traditional diagnostic methods rely on the manual interpretation of chest radiograph images by radiologists, which can be time-consuming and prone to human error, especially when handling large volumes of data.

Furthermore, most existing automated systems focus only on classification (Pneumonia vs Normal) and do not provide additional insights such as severity analysis or interpretability. This limits their practical usability in real world healthcare scenarios.

To address these challenges, this project proposes an AI-based pneumonia detection system that integrates deep learning models with severity analysis and Grad CAM visualization, having the aim to improve both diagnostic accuracy and interpretability.

3.2 Proposed Methodology

The proposed system follows a multi-stage approach that combines image preprocessing, deep learning-based classification, and explainability techniques.

3.2.1 Severity Analysis Mechanism

In addition to pneumonia detection, the proposed system incorporates a severity analysis module to estimate the extent of infection in chest X-ray images. This approach is inspired by the

Radiographic Assessment of Lung Edema (RALE) scoring system, which is widely used in clinical practice to quantify the severity of lung involvement in chest radiographs [7].

The RALE score provides a standardized method for assessing the extent and density of lung opacities, which are indicative of disease severity. Inspired by this concept, the proposed system maps prediction confidence values from deep learning models to severity levels.

The severity classification used in this system is defined as follows:

Table 3.1: Severity Scoring based on RALE-inspired Mapping

Score Range	**Severity Level**
0	Normal
1–2	Mild
3–4	Moderate
5–8	Severe

This mapping allows the system to categorize pneumonia cases into clinically meaningful severity levels. Although the implementation is simplified compared to full RALE scoring, it provides a practical approximation that enhances the interpretability of model predictions.

By integrating severity estimation with classification and Grad-CAM visualization, the system provides a more comprehensive diagnostic output, supporting better clinical decision-making.

The severity classification is defined as follows:

Table 3.2: Severity Scoring Table

Percentage Range	**Score**	**Severity**
$< 5\%$	0	Normal
5% – 24.99%	1	Mild
25% – 49.99%	3	Moderate
50%+	6	Severe

This scoring mechanism helps categorize pneumonia cases into Normal, Mild, Moderate, and Severe levels, making the system more useful for real-world medical applications.

The output shown in Figure 3.1 illustrates the model predictions along with confidence scores and corresponding severity levels. It also demonstrates how different models (ResNet18, DenseNet121, and VGG16) contribute to the final severity estimation.

--- Model Prediction Summary for Sample Image ---

	Model	Prediction	Confidence	Category	Score
0	ResNet18	Normal	75.80%	Severe	6
1	DenseNet121	Normal	54.85%	Severe	6
2	VGG16	Pneumonia	55.00%	Severe	6

--- Severity Scoring Table for Reference ---

	Percentage Range	Score	Severity
0	< 5%	0	Normal
1	5% - 24.99%	1	Mild
2	25% - 49.99%	3	Moderate
3	50%+	6	Severe

--- Analysis of Model Predictions ---

Note: The severity scoring is inspired by concepts from the RALE Score paper.

Figure 3.1: Severity Analysis Output with Model Predictions

1. Data Collection: Chest X-ray datasets are collected from publicly available sources such as Kaggle and the RSNA Pneumonia Detection dataset.

2. Data Preprocessing: The collected images are preprocessed by resizing, normalization, and removal of corrupted data. Data augmentation techniques such as rotation, flipping, and scaling are applied to improve model generalization.

3. Model Development: Pre-trained Convolutional Neural Network (CNN) models are used with transfer learning:

- ResNet18
- DenseNet121
- VGG16

These models are fine-tuned on the dataset to classify chest X-ray images as Pneumonia or Normal.

4. Training and Evaluation: The models are trained using the Adam optimizer and CrossEntropyLoss function. Performance is evaluated using metrics such as accuracy, precision, recall, and F1 score.

5. Explainability using Grad-CAM: Grad-CAM (Gradient-weighted Class Activation Mapping) is applied to generate heatmaps that highlight important regions in the Xray images influencing model predictions.

6. Severity Analysis: A severity scoring mechanism is implemented based on prediction confidence. The output is categorized into:

- Normal
- Mild
- Moderate
- Severe

This module enhances the interpretability of the system by providing not only classification results but also an estimation of disease severity, which is critical for medical decision-making.

3.3 System Design

The proposed system architecture is illustrated in Figure 3.2. The system processes chest X-ray images through multiple stages, including preprocessing, feature extraction through CNN models, and classification.

The system is composed of the following modules:

1. Input Module: Receives chest X-ray images for analysis.

2. Preprocessing Module: Performs resizing, normalization, and data augmentation.

3. Feature Extraction Module: Uses deep learning models to extract meaningful features from images.

4. Classification Module: Classifies images as Pneumonia or Normal using trained CNN models.

5. Explainability Module: Generates Grad-CAM heatmaps to highlight affected regions.

6. Severity Analysis Module: Assigns severity levels based on prediction confidence.

3.4 Details of Hardware/Software Requirements

The implementation of the proposed system requires the following hardware and software components:

Hardware Requirements:

- Processor: Intel i5 or equivalent (or higher recommended)
- RAM: Minimum 8 GB (higher memory preferred for model training)
- Storage: At least 10 GB of available disk space

Software Requirements:

- Programming Language: Python
- Libraries: PyTorch, TensorFlow, NumPy, Pandas
- Development Environment: Google Colab or Jupyter Notebook
- Visualization Tools: Matplotlib, Seaborn

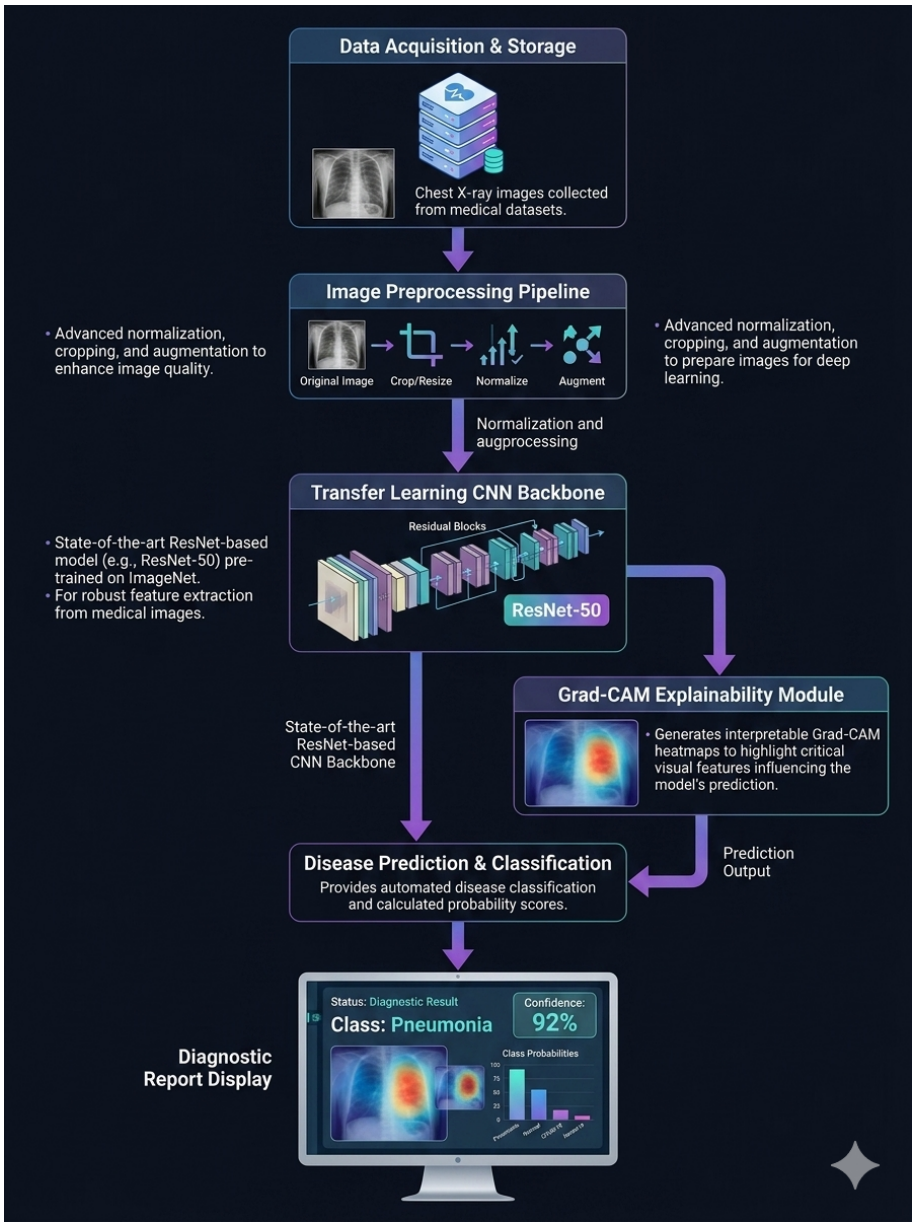

Figure 3.2: Proposed System Architecture

Chapter 4

Results and Discussion

4.1 Implementation Details

The implementation of the pneumonia detection system involved the use of Python within the Google Colab platform. Preprocessing of chest x-rays involved operations such as resizing, normalization, and data augmentation such as rotations and flipping.

In terms of transfer learning, the process involved the use of pre-trained deep learning models such as ResNet18, DenseNet121, and VGG16 to classify whether an image is either Pneumonia or normal.

4.2 Testing Strategy

To test the performance of the models developed, the dataset was separated into train and test data. The following parameters were used for assessing the performance of the models:

- Accuracy
- Precision
- Recall
- F1-score

4.3 Result Analysis

The performance of the models is summarized in Table 4.1.

Table 4.1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
ResNet18	0.88	0.87	0.85	0.86
DenseNet121	0.76	0.74	0.77	0.75
VGG16	0.82	0.80	0.83	0.82

From the above table, it can be seen that ResNet18 is more accurate compared to other two models. Following this model in order of accuracy is VGG16 and DenseNet121.

4.3.1 Model Prediction and Confidence Analysis

Further, for testing the performance of models developed, the prediction output and confidence levels of various models are analyzed comparatively. For performing such comparison, various chest x-ray images are selected as the testing dataset for all the models.

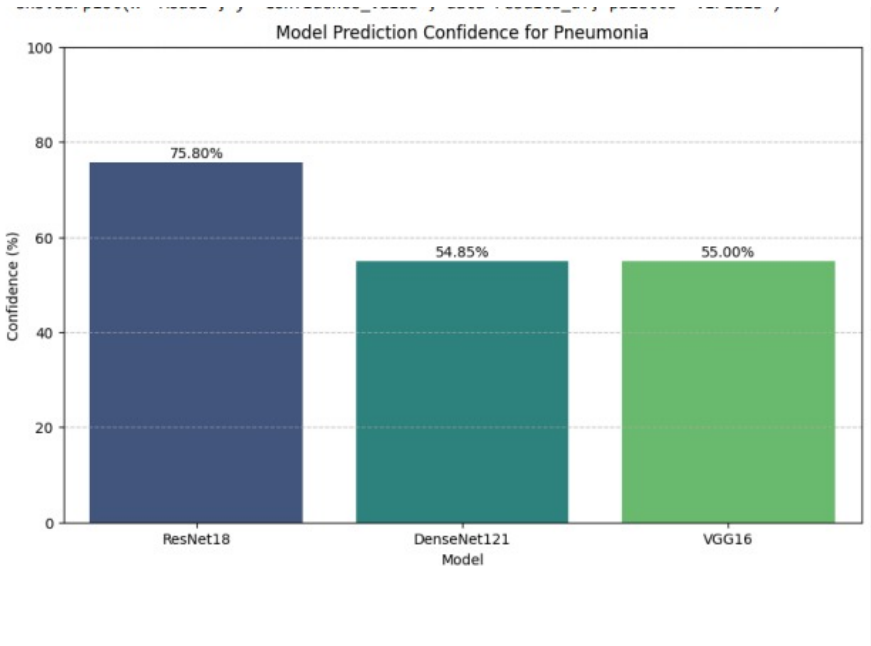

Figure 4.1: Model Prediction and Confidence Comparison

Figure 4.1 shows the predictions made by various models, together with their confidence scores. It can be seen that ResNet18 gives much better prediction results than DenseNet121 and VGG16.

From the above analysis, it can be seen how important it is to use confidence-based criteria to evaluate deep learning models.

4.4 Confusion Matrix Analysis

The confusion matrix is an essential evaluation metric used to analyze the classification performance of deep learning models. It provides detailed insights into correct and incorrect predictions across different classes.

4.4.1 ResNet18

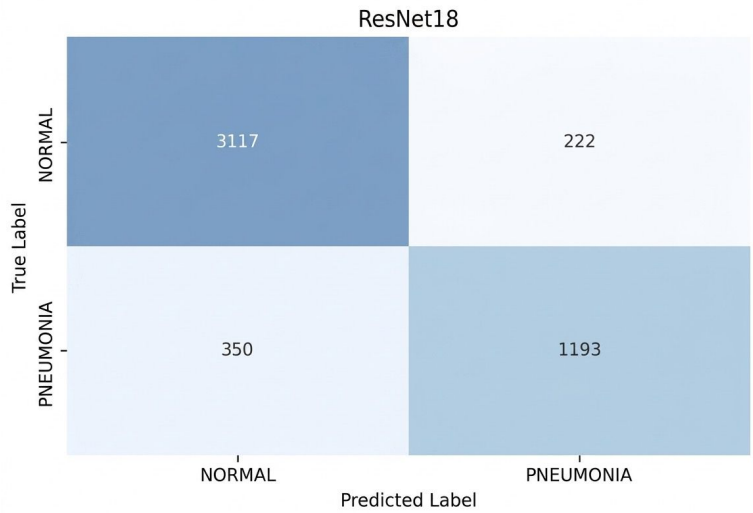

Figure 4.2: Confusion Matrix for ResNet18

The confusion matrix for ResNet18 shows that the model correctly classifies a majority of the samples with minimal misclassifications. This indicates strong discriminative capability and robustness in distinguishing between pneumonia and normal cases.

4.4.2 DenseNet121

DenseNet121 exhibits comparatively higher misclassification rates. In particular, the model struggles in correctly identifying normal cases, which negatively impacts its overall accuracy and reliability.

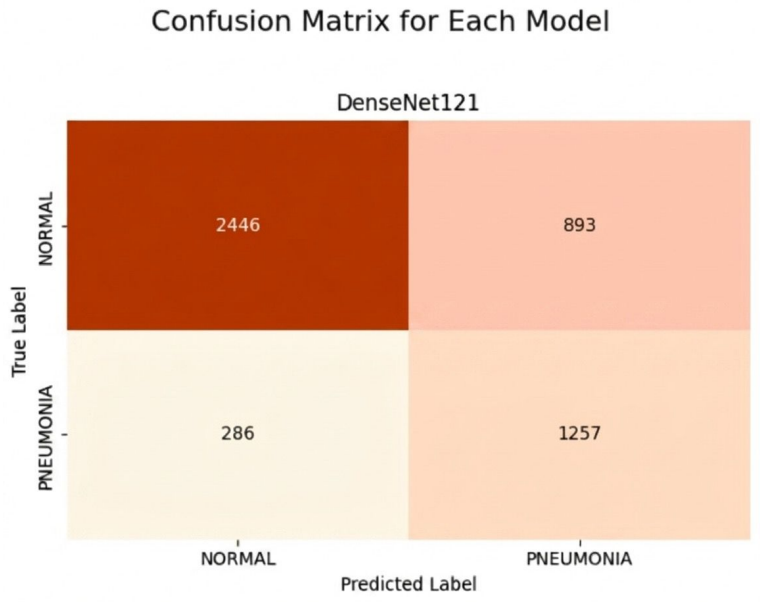

Figure 4.3: Confusion Matrix for DenseNet121

4.4.3 VGG16

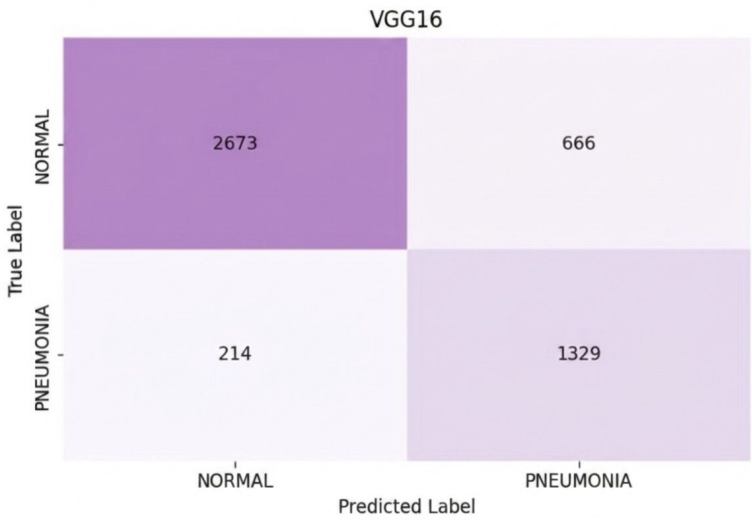

Figure 4.4: Confusion Matrix for VGG16

VGG16 demonstrates moderate performance. Although it performs better than DenseNet121, it still shows a higher number of misclassifications compared to ResNet18.

4.5 Training Accuracy Comparison

The training accuracy of the evaluated models is summarized below:

- ResNet18: 87.63%
- DenseNet121: 77.51%
- VGG16: 81.87%

Among the three models, ResNet18 achieves the highest training accuracy which indicates better learning capability. DenseNet121 records the lowest accuracy, while VGG16 provides moderate performance.

4.6 Overall Model Comparison

A comparative analysis of all models based on accuracy, precision, recall, and F1-score indicates that ResNet18 outperforms the other architectures. It demonstrates strong generalization ability with minimal overfitting.

DenseNet121 shows the weakest performance among the three models, while VGG16 achieves moderate results but does not match the effectiveness of ResNet18.

Therefore, ResNet18 is selected as the optimal model for pneumonia detection in this study.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

In summary, this project developed an intelligent pneumonia diagnosis system based on machine learning, which involved the application of severity analysis and Grad-CAM visualization methods to enhance the performance and interpretability of the medical imaging system.

As described earlier, various deep learning algorithms including ResNet18, DenseNet121, and VGG16 were used in the proposed system through the use of transfer learning. From the experimental results, it can be seen that the ResNet18 model exhibited a higher accuracy rate than other tested models, hence its superior performance in the experiments.

Additionally, the integration of severity analysis and Grad-CAM visualization in the proposed pneumonia detection system provided important insights concerning model prediction and improved its interpretability. As shown in the experimental results, the Grad-CAM method indicated the affected regions in chest X-ray images by highlighting their locations.

From the above discussion, it can be concluded that the use of deep learning models and interpretability technique led to an advanced and reliable pneumonia detection method. Thus, the goals of this research project have been accomplished satisfactorily.

Limitations:

However there are some shortcomings with the system despite its high performance. One of these is that model performance is highly dependent upon the quality and variability of the data set. In addition, the severity of the problem is assessed based on the confidence of predictions which may not be entirely compatible with medical guidelines.

Another problem is the amount of computational power needed for training the deep learn-

ing models. Moreover, the proposed system lacks a complete user interface.

Real-World Applicability:

This system can be implemented in hospitals, laboratories, and telemedicine systems to aid health care practitioners in diagnosing pneumonia at an early stage.

The addition of the Grad-CAM visualization will make this system useful for practical applications as the transparency of predictions will help in gaining trust among the users in medical fields.

5.2 Future Scope

Even though the suggested algorithm demonstrates high accuracy in classifying images, several possible improvements can be considered. Advanced algorithms like EfficientNet and transformers can be considered to boost its efficiency.

Moreover, the algorithm may be expanded through the use of large-scale databases with various types of images to enhance the system's ability to cope with new cases and provide more accurate results.

Furthermore, the model can be upgraded through the creation of an online application that will allow the user to input an image and receive an immediate result. In such a way, the algorithm will become more convenient.

Moreover, the analysis of disease severity can be made more accurate through the incorporation of clinical parameters and professional annotation.

Finally, the system can be adapted to identify other pulmonary diseases such as tuberculosis and lung cancer.

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Appendices

Appendix A

Weekly Progress Report

STAGE	DATES	WORK DONE
Problem Definition and Dataset Collection	Feb 11 – Feb 17	Identified pneumonia detection problem, collected and merged datasets, and finalised project scope.
Literature Survey	Feb 18 – Feb 24	Reviewed existing automated diagnostic methods for pneumonia detection and identified gaps in knowledge.
Data Preprocessing and Augmentation	Feb 25 – Mar 2	Preprocessed chest X-ray images, applied data augmentation techniques such as rotation and
Model Development (Initial Training)	Mar 3 – Mar 16	Fine-tuned pre-trained CNN models (ResNet18, DenseNet121, VGG16) on collected dataset.
Severity Analysis Implementation	Mar 17 – Mar 23	Fine-tuned pre-trained CNN models (ResNet18, DenseNet121, VGG16) on collected dataset.
Grad-CAM Visualization	Mar 24 – Mar 30	Implemented severity analysis module to classify pneumonia cases into mild, moderate, and severe categories.
Grad-CAM Visualization	Mar 24 – Mar 30	Integrated Grad-CAM visualization to generate heatmaps highlighting critical regions in X-ray images.
Evaluation and Report Finalization	Mar 31 – Apr 5	Evaluated models using accuracy, precision, recall, F1-score, and finalized project report.

Mini Project Weekly Progress Report

Figure A.1: Weekly Progress Report

Appendix B

Plagiarism Report

Plagiarism report will be attached during final submission.

Appendix C

Publication Details / IPR

1. AI-based Pneumonia Detection with Severity Analysis and Grad-CAM (Manuscript prepared for submission to a journal/conference.)

Appendix D

Project Competitions

Acknowledgments

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Date: _____